

Verifier-Grounded Evaluation of LLM-Generated Network Configurations: A Preregistered NetConfig-Semantic Study

Autonomous AI Researcher
CNSM 2026 Agentic AI Researcher Track

Abstract—We report a fully preregistered, artifact-anchored paired experiment (CAND-2026-001-NetConfig-Semantic-v1; preregistration SHA-256 7db1663cf3982c8ea1e16c3dd7c0348f356bb4cbe3978ecc75bb59fdebb71a37) comparing a deterministic template baseline to a single-pass LLM generator (declared_model_version = gpt-5-mini) on $N = 120$ synthetic intent→configuration tasks using a deterministic validator. Under the frozen confirmatory semantics (shared_initial_candidate per pair) all confirmatory episodes completed: baseline = 120/120 verifier passes; guarded (gpt-5-mini, seed_0) = 120/120 verifier passes. Paired difference = 0.00; 95% bootstrap percentile CI [0.00, 0.00] (10,000 resamples, seed = 1729); exact McNemar two-sided $p = 1.0$. We (i) publish the exact execution and analysis artifacts and SHA-256 fingerprints needed to reproduce the run (see execution/execution_manifest.json in the artifact bundle), (ii) diagnose—using archived artifacts—why the paired outcome is uninformative about independent generative reliability (preregistered shared_initial_candidate design, template-tight task synthesis, and validator–task coupling), and (iii) provide artifact-grounded, runnable modifications (independent_per_condition sampling, multi-seed confirmatory design, validator diversity, and expanded task heterogeneity) that preserve reproducibility while producing informative independent comparisons. The immutable initial master prompt is archived (provenance/master_prompt.txt; SHA-256 1872df1e1805d2d96940456ca016bd665d1d5196add77f5acdff1582bb39b15ba).

I. INTRODUCTION

Motivation. Translating operator intent into device configuration is a core NetOps task where correctness is safety-critical: incorrect commands can break reachability, violate ACLs, or create unsafe transient states during multi-step changes. Deterministic offline verifiers (reachability/ACL/routing analyzers such as Batfish-class tools) are widely used to validate intended changes before deployment and provide an operational oracle for generated configuration artifacts [https://doi.org/10.1145/3603269.3604866]. Large language models (LLMs) offer rapid intent→config generation but raise reliability questions: how often do independently sampled LLM candidates satisfy operational verifier constraints? How do LLM pipelines compare with deterministic template baselines when correctness is judged by a deterministic validator?

Research question and contribution. After a fresh autonomous literature and gap analysis we preregistered (CAND-2026-001-NetConfig-Semantic-v1) a paired confir-

matory test asking: does a single-pass LLM generator (gpt-5-mini) have a lower verifier pass rate than a deterministic template baseline across $N = 120$ intent→config tasks? Our primary contribution is an artifact-anchored, fully reproducible preregistered execution + analysis bundle and a transparent diagnosis of why the preregistered confirmatory semantics (shared_initial_candidate) produced a degenerate but correct paired outcome. We (1) publish the exact artifacts and SHA-256 fingerprints required for byte-for-byte reruns; (2) report confirmatory counts and preregistered statistical inferences grounded in the archived traces (baseline = 120/120, guarded = 120/120; paired difference = 0.00; bootstrap CI [0.00,0.00]; McNemar $p = 1.0$); and (3) provide artifact-grounded, immediately runnable experimental modifications that preserve reproducibility while answering the operational question of independent generative reliability.

II. RELATED WORK

Verifier-grounded NetOps evaluation and intent→config generation. Offline semantic validators and configuration analyzers (Batfish and successor research) are a mature operational pattern for pre-deployment correctness checks; they compute forwarding/reachability and detect ACL or routing policy violations from static device configurations, and have been demonstrated to scale to realistic networks [Brown et al., 2023; doi:10.1145/3603269.3604866]. Work that couples generative systems to such verifiers typically adopts a generate–validate–repair pattern: a generator proposes one or more candidates, a verifier checks deterministic semantic properties, and a repair loop attempts to fix failing candidates. That architectural pattern is the most direct route to operationally meaningful automation because it separates fluent text generation from deterministic, auditable correctness checks.

What prior LLM→NetOps evaluations vary on. Recent intent→configuration studies differ along three methodologically important axes: (A) sampling semantics (single canonical candidate per task vs. independent per-condition or multi-seed sampling), (B) repair budget and loop semantics (single-pass, bounded K iterations, or unbounded repair), and (C) validator scope (syntactic parsing only, reachability/ACL/routing checks, or multi-verifier ensembles). For example, several practical systems and empirical studies emphasize iterative repair and small fixed repair budgets, with empirical evidence that most repair gains arise in the first three iterations

(see "Is Three the Magic Number? An Empirical Evaluation of LLM-Based Repair Loops"; arXiv/openalex record W7167748939). Those prior evaluations that used independent per-condition sampling and multi-seed replication address the operational reliability question—i.e., the probability that independent LLM outputs satisfy verifier constraints—directly, at the cost of more compute and artifact volume.

How verifier choice and task synthesis shape measured reliability. Two patterns recurring in the literature and in our artifact corpus are (i) template-aligned tasks that encode explicit `required_sequences` and per-task `workflow_policies` (e.g., `must_be_down_for_fields`, `minimum_active_in_groups`) which a deterministic validator checks exactly, and (ii) reuse of a single canonical candidate across multiple scoring episodes. When tasks are generated from tight templates, a generator that reproduces the template pattern will often pass a deterministic verifier even if it would fail on more open, paraphrased intents; conversely, independent sampling exposes generation variance and highlights failure modes. This methodological dependence is central: a verifier can only assess the artifact it is given, and design choices that reduce generation variance (e.g., using a `shared_initial_candidate` across conditions) will necessarily collapse paired contrasts that aim to measure independent generative reliability.

Artifact-anchored reproducibility as a scientific stance. A recurring shortcoming in several prior studies is incomplete artifact publication (missing seeds, missing provider traces, or missing per-episode validator traces), which obstructs exact reruns and post-hoc diagnosis. Our work purposely publishes exhaustive artifacts—including model provider call traces, the canonical shared responses, and the per-episode verifier JSON traces—so that any reader can deterministically reconstruct the exact causal chain that produced the reported statistics. Representative artifacts we publish and that directly support our diagnosis are: `execution/responses/task-000001-shared-initial.txt` (SHA-256 `8f38c8becd7e1e36...`), the per-episode validator trace `execution/scoring/task-000001-guarded.json` (SHA-256 `0d56c1add7c5a57f...`), and the full execution manifest `execution/execution_manifest.json` which lists every file and SHA-256 used in the run. These exact references permit external auditors to verify that identical candidate inputs were scored across both conditions (the core reason the paired contingency collapsed to $n_{11} = 120$ in our run).

Empirical and methodological contrasts with prior work. Many prior evaluations that report conditional pass probabilities adopt independent sampling strategies (separate seeds per condition) or explicitly measure seed sensitivity across multiple draws to estimate the LLM’s stochastic reliability. Studies emphasizing iterative bounded repair budgets and repair-loop dynamics also show that orchestration and feedback design often dominate raw model choice for repair effectiveness (see the iterative-repair literature referenced in our evidence bundle). By contrast, our preregistered confirmatory design chose `shared_initial_candidate` semantics to control for candidate content while isolating the validator behavior; that design answers a different estimand (validator consistency

under identical inputs) and therefore must be interpreted differently than independent-sampling studies. We explicitly contrast these approaches here to make clear what inference the archived confirmatory statistics legitimately support and what they do not.

Contamination detection and disclosure in the literature. The concern that a generator’s proposals may match training data (pretraining exposure) is recognized across the LLM evaluation literature. We preregistered and ran contamination heuristics (exact-match and normalized n-gram overlap) and recorded the outputs in `analysis/contamination_summary.csv`; however, string-overlap heuristics are imperfect proxies for training exposure, so the absence of a flag is not definitive proof of zero exposure. Publishing provider call traces and canonical candidate hashes (the files and SHA-256 fingerprints above) permits more exhaustive external contamination analyses without changing the original experimental artifacts.

Synthesis: what an evidence-anchored `related_work` tells practitioners. The practical lesson across the verifier-grounded literature is twofold and operationally salient: (1) rigorous verifier grounding is necessary to obtain auditable, safety-relevant pass/fail judgements for generated configurations, and (2) experimental generation semantics must match the operational question of interest—shared canonical candidates test validator determinism and scoring reproducibility; independent per-condition sampling and multi-seed designs estimate generative reliability under deployment-like stochasticity. Our artifact bundle is intentionally complete so that other researchers can replay either paradigm (`shared_initial` or `independent_per_condition`) deterministically and thereby answer the specific operational question they care about without ambiguity.

III. METHODOLOGY

Preregistration and frozen design. We preregistered study

CAND-2026-001-NetConfig-Semantic-v1 prior to observing any model outputs; the preregistration artifact is archived (SHA-256 `7db1663cf3982c8ea1e16c3dd7c0348f356bb4cbe3978ecc75bb59fdded71a3`). The preregistration specified: paired confirmatory design across $N = 120$ tasks (40 tasks per topology class: edge, datacenter, multi-AS), primary estimand `paired_success_rate_difference_guarded_minus_baseline`, confirmatory `generation_semantics` = "`shared_initial_candidate`", initial canonical seed_0 per pair, validator = `deterministic_netops_validator_v5`, and the exact analysis plan (`paired_binary_analysis_v1`) including bootstrap parameters (10,000 resamples, seed = 1729). The full execution manifest and per-file SHA-256s are published (`execution/execution_manifest.json`; see artifact manifest in submission bundle).

Task synthesis and templates. Tasks were programmatically synthesized from a deterministic template bank (`generator_id` `deterministic_netops_task_generator_v5`, version 5.0). Each task manifest entry encodes: `initial_state`, `expected_final_state`, an explicit `required_sequence` (the minimal change

sequence that must be executed to satisfy intent safely), per-task workflow_policies (e.g., must_be_down_for_fields, minimum_active_in_groups), allowed_touched_fields, and a difficulty annotation (changed_interface_count, transient_rule_count). Representative task manifest entries and their SHA-256s are archived (e.g., execution/task_manifest.jsonl; see task examples under artifact_examples). Templates intentionally encode operationally meaningful transient constraints (e.g., make-before-break switchover, atomic MTU changes) so that verifier checks carry operational semantics rather than purely syntactic checks.

Model, orchestration, and generation semantics. The declared LLM was gpt-5-mini accessed via the hosted adapter family hosted_netops_gvr_v1 (execution/model_configuration.json; SHA-256 a40e96e212ab87da251ac0d6dbb75bc543afa13438b5a6b9ebd0735077ad820b). The preregistered confirmatory generation semantics produced a single canonical candidate per pair (shared_initial_candidate) and reused that identical candidate for both baseline and guarded scoring episodes; the canonical candidate files are archived at execution/responses/*-shared-initial.txt (representative SHA-256s: task-000001 shared initial SHA-256 8f38c8becd7e1e36..., task-000002 SHA-256 ae967f70dfb6ccb8..., task-000003 SHA-256 f7f4db249ecbbce...).

Deterministic validator and scoring. Each candidate was scored by deterministic_netops_validator_v5 (validator_version 5.0). The validator parses commands, normalizes configuration sequences, enforces per-task workflow_policies, runs transient safety checks (e.g., minimum_active_in_groups during the change), and computes final_state reachability and ACL invariants. The scoring rule used for the confirmatory test was full_intent_constraint_validation: a binary pass (VALID_CONFIGURATION = 1) requires that all required_commands and transient constraints are satisfied. Each scoring episode emits a detailed JSON trace archived at execution/scoring/*.json (representative: execution/scoring/task-000001-guarded.json SHA-256 0d56c1add7c5a57f...).

Execution accounting, retries, contamination heuristics. Planned_episode_count = 240 (120 pairs $\times$ 2 conditions). Completed_episode_count = 240; model_calls_used = 120 (shared_initial generation per pair counted once); failed_episode_count = 0. A preregistered retry policy allowed up to two automated retries for infrastructure failures; none were required. Contamination heuristics (exact-match and normalized n-gram overlap) were run as preregistered and recorded (analysis/contamination_summary.csv). All provider call traces and call_cache artifacts are included (execution/provider_calls/*.json, execution/call_cache/*) to permit external exposure analysis.

IV. RESULTS

Confirmatory counts and inferential summary (artifact grounding). All confirmatory scoring episodes completed and are traceable in the execution manifest

(execution/execution_manifest.json). The archived confirmatory counts are: baseline_success_count = 120/120; guarded_success_count = 120/120; complete_pair_count = 120. The paired contingency table is degenerate: $n_{11} = 120$, $n_{10} = 0$, $n_{01} = 0$, $n_{00} = 0$ (analysis/paired_contingency_table.csv SHA-256 9f25790c7eeaafb3...). The primary estimator (mean of guarded_i - baseline_i) = 0.00. The preregistered bootstrap (10,000 resamples, seed 1729) yields a 95% percentile CI [0.00, 0.00]; exact McNemar two-sided $p = 1.0$. Analysis artifacts (condition_summary.csv SHA-256 9c77c96f37c4b6ff..., paired_difference_figure.svg SHA-256 ae5b8e0c644f8cf7...) are published in the bundle.

Representative validator traces (artifact examples). The per-episode JSON traces record parsed_command_count, required_command_count, normalized_configuration, and violation_count. Examples: execution/scoring/task-000001-guarded.json (SHA-256 0d56c1add7c5a57f...) shows parsed_command_count = 4, required_command_count = 4, violation_count = 0 and contains explicit before/after final_state mappings. The canonical candidate used for that pair is archived at execution/responses/task-000001-shared-initial.txt (SHA-256 8f38c8becd7e1e36...). Multiple pairs exhibit identical baseline/guarded/shared response fingerprints (e.g., task-000003 artifacts all share SHA-256 f7f4db249ecbbce...), demonstrating that identical candidates were evaluated across both conditions.

Why the confirmatory outcome is degenerate (artifact-grounded diagnosis). The degenerate contingency (all pairs n_{11}) follows directly from two preregistered, artifact-visible design choices: (1) the confirmatory generation_semantics set to shared_initial_candidate caused a single canonical candidate to be reused for both baseline and guarded episodes per pair (see execution/responses/*-shared-initial.txt SHA-256s), and (2) the task templates encoded explicit required_sequences and workflow_policies that canonical candidates satisfied. Under these conditions a deterministic validator necessarily returns identical pass/fail judgements for identical inputs, hence the paired difference must be zero. The execution and scoring artifacts (execution/responses/* and execution/scoring/*) fully document this chain of causation and are published to enable exact reruns.

Contamination and exposure. Preregistered contamination heuristics flagged no confirmatory items under the chosen thresholds (analysis/contamination_summary.csv). All provider call traces and call_cache entries are published to permit stronger external exposure checks; absence of a flag in the preregistered heuristic is not proof of zero pretraining exposure and is explicitly noted in the Disclosure.

V. DISCUSSION

Interpretation of confirmatory inference. The confirmatory analysis correctly reports that, for the preregistered estimand and generation semantics, the deterministic

verifier returned identical pass judgments for both conditions: `baseline_success_count` = 120/120 and `guarded_success_count` = 120/120, producing a paired difference of 0.00 with a degenerate 95% bootstrap percentile CI [0.00,0.00] (10,000 resamples, seed = 1729) and exact McNemar p = 1.0. This numerical outcome is a direct and reproducible consequence of two archived facts: (A) the preregistered `generation_semantics` = "shared_initial_candidate" produced one canonical candidate per task that was reused for both the baseline and guarded episodes (see `execution/responses/*-shared-initial.txt`; representative SHA-256s include `task-000001: 8f38c8becd7e1e36...`, `task-000002: ae967f70dfb6ccb8...`, `task-000003: f7f4db249ecbbce...`), and (B) the deterministic validator (`deterministic_netops_validator_v5`) is idempotent on identical inputs and enforces the template-encoded `required_sequences` and transient safety constraints exactly (see `execution/scoring/*.json`; representative SHA-256 `execution/scoring/task-000001-guarded.json` `0d56c1add7c5a57f...`). Because both conditions received byte-identical canonical candidates, identical validator outputs were the only logically possible result; the archived `execution_manifest` records `model_calls_used` = 120 while `planned_episode_count` = 240, confirming the single-generation per pair accounting (`execution/execution_manifest.json` SHA-256 `7db1663cf...` in preregistration and manifest entries).

What this confirms—and what it does not. The run validates the preregistered estimand (whether the verifier treats identical inputs differently across scoring conditions) and demonstrates complete reproducibility through the published artifact manifest and per-file SHA-256s. It does not, however, estimate independent LLM generative reliability (the per-task probability that an independently sampled LLM candidate satisfies the verifier). That latter operational quantity requires independent per-condition sampling (distinct generation seeds or independent generation calls per condition), which the archived manifest shows was not performed for the confirmatory test (`model_calls_used` = 120 vs. the 240 episodes scored). Interpreting the observed 120/120 pass rates as evidence that independently sampled LLM outputs will pass the validator at the same rate would therefore be a category error; the artifacts make the provenance of this limitation transparent.

How archived artifacts enable immediate, rigorous reruns. One virtue of full artifact publication is that the minimal changes needed to answer the operational question are runnable and deterministic. Using the provided bundle (`execution/responses`, `execution/provider_calls`, `execution/call_cache`, `execution/scoring`, `execution/task_manifest.jsonl`, and `execution/model_configuration.json`), researchers or practitioners can (without changing task templates or the validator) (i) switch to `independent_per_condition` semantics (generate a distinct candidate per condition per task) and reexecute scoring; this deterministically increases `model_calls_used` from 120 toward the preregistered maximum of 240 and breaks the shared-input coupling that produced the degenerate contingency, or (ii) adopt a multi-seed confirmatory design ($K \geq$

3 independent seeds per condition) to estimate per-condition pass probabilities with bootstrap CIs. Both reruns preserve byte-for-byte reproducibility for all unchanged artifacts because the original provider traces and call cache are published (`execution/provider_calls/*.json` and `execution/call_cache/*`). The manifest documents exact file paths and SHA-256s so external auditors can verify that only the generation semantics changed while all other artifacts remain identical.

Validator diversity and construct validity. The deterministic validator enforces the template-expressed `workflow_policies` exactly; this is an asset for operational safety but a construct-validity risk for generalization because template-aligned canonical candidates are more likely to match validator expectations. The archived scoring traces quantify this coupling (`parsed_command_count`, `required_command_count`, and the `validation_before/validation_after` `final_state` in `execution/scoring/*.json`). A pragmatic mitigation—supported by the archived bundle—is to add a second independent validator (e.g., a reachability emulator or an alternate static analyzer) and report both intersection and union pass rates. Because validator outputs are archived as JSON traces, adding a validator is an orthogonal rerun that preserves the original artifacts while increasing construct validity of the decision rule.

Threats to validity and how they are addressed by the artifacts. Internal validity: the `shared_initial` design intentionally removed generation variance and thus precluded testing independent generative reliability; this is visible in the manifest (`model_calls_used` = 120) and in the repeated `shared_initial` SHA-256 fingerprints. Construct validity: the task templates' explicit `required_sequences` and `workflow_policies` increase the probability that a canonical, template-aligned candidate will pass the validator; the task manifest entries (`execution/task_manifest.jsonl`) and per-task payloads in the bundle document these constraints and make the construct limitation auditable. External validity: the study used a single declared model (gpt-5-mini) and synthetic, template-generated tasks; all such scope limitations are preserved in the preregistration and manifest so consumers can judge applicability. Conclusion validity: the degenerate contingency yields exact but uninformative inference about independent generative reliability; the archived bootstrap parameters (10,000 resamples, seed = 1729) and generated bootstrap artifact (`analysis/paired_difference_figure.svg`) ensure this conclusion is verifiable.

Operational implications for NetOps. From an operator perspective the artifact-anchored diagnosis yields three practical rules grounded in the archived evidence: (1) Verifier grounding is necessary but insufficient—experimental generation semantics must match the deployment question. The manifest proves that identical inputs produce identical verifier outputs; reuse of canonical candidates therefore invalidates claims about independent sampling. (2) Reproducible preregistration + artifact publication materially reduces ambiguity about what a confirmatory test measures and accelerates corrective reruns

that answer operational questions (the exact rerun steps are fully executable from the published bundle). (3) For deployment decisions, practitioners should report per-condition pass probabilities estimated under independent sampling and, when possible, validate across multiple independent validators; both methodological changes are artifact-supported reruns rather than new experiments and preserve full reproducibility.

Concluding remark. The confirmatory run executed exactly as preregistered and the archive clearly documents why the observed paired outcome is degenerate for the operational quantity of independent LLM reliability. That transparent, artifact-level diagnosis is the central scientific value of this submission: it shows how a carefully preregistered, fully archived pipeline can expose subtle estimation/semantics mismatches and enable deterministic, reproducible corrective reruns that answer the operational questions NetOps teams require.

VI. LIMITATIONS

- Controlled synthetic tasks: programmatic templates produced narrow required_sequences and workflow_policies that favour canonical solutions and limit external generalizability to heterogeneous operational intents.
- Confirmatory shared_initial semantics: the preregistered generation_semantics = 'shared_initial_candidate' supplied identical canonical candidates to both conditions per pair, reducing independence between conditions and causing the degenerate paired outcome.
- Single canonical seed for confirmatory test: the primary analysis used one canonical seed per task; preregistered exploratory multi-seed analyses (seeds_1..4) were not included in the confirmatory inference reported here.
- Validator-task coupling: the deterministic validator enforces constraints encoded in the task templates, which can increase pass rates for template-aligned candidates and reduce construct validity for open distributions.
- Possible training-data exposure: preregistered contamination heuristics found no flags under chosen thresholds, but absence of a flag is not proof of zero exposure to related artifacts in model pretraining.
- Single-model, single-validator run: results apply only to gpt-5-mini under the recorded configuration and deterministic_netops_validator_v5; they do not generalize across model families, agentic pipelines, or other validators.

VII. CONCLUSION

We executed a fully preregistered, verifier-grounded paired experiment and released a complete artifact bundle enabling byte-level reproduction (execution/execution_manifest.json and analysis/*). Under the preregistered confirmatory semantics (shared_initial_candidate) both the deterministic template baseline and the gpt-5-mini guarded condition validated on all N = 120 tasks (both 120/120). Archived evidence (shared_initial_candidate files and validator scoring traces) demonstrates that identical canonical

candidates per pair and template-tight task constraints caused the degenerate paired outcome; consequently the confirmatory paired result does not provide evidence about independent LLM generative reliability in operational settings. We provide artifact-grounded, runnable modifications (independent_per_condition sampling, multi-seed confirmatory execution, validator diversification, task heterogeneity expansion) that preserve reproducibility and enable informative independent comparisons. All execution and analysis artifacts—including the immutable master prompt reference (provenance/master_prompt.txt; SHA-256 1872df1e1805d2d96940456ca016bd665d1d5196add77f5acdf1582bb39b15) and the preregistration SHA-256—are included in the submission bundle to permit exact audit and follow-up experiments. Transparent preregistration combined with exhaustive artifact publication clarifies precisely what a confirmatory test measures and accelerates corrective reruns that answer the operational questions NetOps practitioners require for deployment decisions.

DISCLOSURE STATEMENT

Mandatory disclosure (CNSM Agentic AI Researcher track). LLM(s) and provider: declared_model_version = gpt-5-mini accessed via provider adapter 'openai_responses' and adapter family 'hosted_netops_gvr_v1' (execution/model_configuration.json SHA-256 a40e96e212ab87da251ac0d6dbb75bc543afa13438b5a6b9ebd073597ffdd82). Orchestration/framework: hosted_netops_gvr_v1 executed the preregistered pipeline; deterministic scoring used deterministic_netops_validator_v5 (validator_version 5.0). Preregistration: CAND-2026-001-NetConfig-Semantic-v1 (frozen prior to execution); preregistration SHA-256 7db1663cf3982c8ea1e16c3dd7c0348f356bb4cbe3978ecc75bb59fddb71a3. Exact initial master prompt: preserved as an immutable archived file provenance/master_prompt.txt (SHA-256 1872df1e1805d2d96940456ca016bd665d1d5196add77f5acdf1582bb39b15). Human role and autonomy boundary: the Research Director selected the topic family (Generative AI and LLMs for NetOps) and approved the initial master prompt before the autonomy lock. After the autonomy lock the autonomous pipeline performed literature review, research-gap selection, preregistration, code generation, experiment execution, deterministic validation, statistical analysis, autonomous internal peer review, manuscript drafting and revision. No human manually edited technical manuscript content after the autonomy lock. Preregistered confirmatory generation semantics and consequence: confirmatory generation_semantics was set to 'shared_initial_candidate' so each pair received an identical canonical candidate for both baseline and guarded episodes; representative shared candidate artifacts: execution/responses/task-000001-shared-initial.txt (SHA-256 8f38c8becd7e1e36...), execution/responses/task-000002-shared-initial.txt (SHA-256 ae967f70dfb6c...), execution/responses/task-000003-shared-initial.txt (SHA-256 f7f4db249ecbbce...). Validator and scoring: deterministic_netops_validator_v5 performed

normalized parsing, intent constraint checking, transient safety checks, and emitted per-episode JSON scoring traces archived under `execution/scoring/*`.json (representative SHA-256s: `execution/scoring/task-000001-guarded.json` SHA-256 `0d56c1add7c5a57f...`). Execution accounting and retries: `planned_episode_count` = 240, `completed_episode_count` = 240, `model_calls_used` = 120, `failed_episode_count` = 0; preregistered retry policy allowed up to 2 automatic retries for infrastructure failures; none were required. Contamination checks and reproducibility: preregistered string-overlap heuristics found no confirmatory flags under chosen thresholds, but absence of a flag is not proof of zero exposure to related artifacts in model pretraining. All artifacts, seeds, code, and per-file SHA-256 hashes required for exact reproduction are released in the submission bundle (`execution/execution_manifest.json` contains the exhaustive manifest). The initial master prompt is referenced immutably by path and SHA-256 above.

REFERENCES

- [1] <https://doi.org/10.1145/3603269.3604866>
- [2] <https://doi.org/10.23919/cnsm62983.2024.10814407>
- [3] <https://doi.org/10.1002/nem.70029>